Labwork 1

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I. Introduction

This report attempts to recreate the results obtained from the ECG heartbeat classification paper by Mohammed et al.[1] (2018).

The original paper discusses the classification of heartbeats into five different arrhythmias based on the AAMI EC57 standard. It additionally presents a method for knowledge transfer to myocardial infarction. This report limits its scope to the classification of heartbeats without knowledge transfer; therefore, the PTB Diagnostic dataset is not utilized.

The data was obtained from the authors' Kaggle dataset[2]. Consequently, only the processed MIT-BIH Arrhythmia dataset is used in this report.

The implemented model architecture is inspired by the one presented in the reference paper. It was trained using categorical cross-entropy as the loss function, the Adam optimizer, and accuracy as the primary evaluation metric. Training was conducted with a batch size of 32 over 50 epochs.

II. Data Description

The dataset sourced from Kaggle contains processed data derived from the original MIT-BIH dataset[3]. This section strictly describes the processed arrhythmia dataset utilized for this model.

The dataset contains a total of 109,446 samples. The sampling frequency is 125 Hz, and each sample consists of 187 columns. The data is divided into five categories: N (0), S (1), V (2), F (3), and Q (4).

The training set contains 87,554 samples, while the test set contains 21,892 samples. Class 0 (N) is the most common category. Both sets exhibit a highly similar class distribution, which is summarized in Table I.

Class	**Training Set**	**Test Set**
0.0	0.8277	0.8276
1.0	0.0254	0.0254
2.0	0.0661	0.0661
3.0	0.0073	0.0074
4.0	0.0735	0.0735

TABLE I
PROPORTIONAL CLASS DISTRIBUTION IN THE TRAINING AND TEST SETS.

Prior to training, the classes are balanced using the `resample` utility from the `scikit-learn` library.

[1] https://arxiv.org/abs/1805.00794
[2] https://www.kaggle.com/datasets/shayanfazeli/heartbeat
[3] https://physionet.org/content/mitdb/1.0.0/

III. Model Implementation

The implemented model architecture is based on the one proposed in the reference paper. The core component is a residual block consisting of two 1-D convolutional layers (kernel size 5, padding 2) and one max-pooling layer (kernel size 5, stride 2).

Within the block, the input is stored as a residual connection. It is passed through the first convolutional layer followed by a ReLU activation function, and then processed by the second convolutional layer. The stored residual is added to this output, followed by another ReLU activation and the max-pooling layer.

The complete neural network utilizes five such residual blocks. Initially, the input is processed by a 1-D convolutional layer with 32 output channels, a kernel size of 5, and a padding of 2. The output subsequently passes through the five residual blocks (each with 32 channels). Afterward, the output tensor is flattened and proessed by two fully connected layers, each followed by a ReLU activation. Finally, a linear classifier layer outputs the five class features.

IV. Training

The training process utilizes the Adam optimizer and an `ExponentialLR` scheduler to decay the learning rate every 10,000 training iterations. Categorical cross-entropy is employed as the loss function. The model is trained for 50 epochs.

V. Result

The final evaluation on the testing dataset yielded an accuracy of 98.45%.